

# **Bias Collapse in Shallow ReLU Networks**

Numerical Evidence for Open Problems 4.1, 4.2, and 4.3

NSF RTG Project 4

Auburn University, NSF RTG

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# Motivation

## The overparameterization puzzle:

- ▶ Modern networks have far more parameters than the problem demands
- ▶ Classical learning theory predicts overfitting; practice shows robust generalization
- ▶ Why does the optimizer self-simplify?

## Bias collapse: a concrete mechanism

- ▶ Gradient flow automatically eliminates redundant neurons, with no regularization required
- ▶  $m = 1500$  neurons  $\rightarrow$  effective complexity of  $k = 7$  neurons
- ▶ The number of surviving *bias clusters* equals the number of *inflection points* of the target function

### This work

Rigorous numerical investigation of three open problems:

1. **OP 4.1:** Does  $C(m, f^*) \rightarrow k$  as  $m \rightarrow \infty$ ?
2. **OP 4.2:** Does collapse generalize to  $d \geq 2$  dimensions?
3. **OP 4.3:** Is there a provable  $L^2$  pruning bound after collapse?

**72 runs**, 9 targets,  $m$  up to 5000.

# Model and Gradient Flow

1D shallow ReLU network:

$$f(x) = \sum_{j=1}^m a_j \sigma(x - b_j), \quad \sigma = \max(0, \cdot), \quad x \in [-1, 1]$$

Each neuron places a *kink* at bias  $b_j$  with slope change  $a_j$ .

**Gradient flow on MSE**  $\mathcal{L} = \frac{1}{2} \int_{-1}^1 (f - f^*)^2 dx$ :

$$\begin{aligned} \dot{a}_j &= - \int_{-1}^1 (f - f^*) \sigma(x - b_j) dx \\ \dot{b}_j &= a_j \int_{b_j}^1 (f - f^*) dx \end{aligned}$$

Solved via `scipy` RK45,  $T = 500$ ,  $\text{rtol} = 10^{-4}$ .

Stationarity criterion:  $\max_j |\dot{a}_j|, |\dot{b}_j| < 0.01$ .

**Key measured quantity:**

$$C(m, f^*) = \#\{\text{bias clusters}\}$$

at stationarity; gap threshold 0.02.

## Fixed-point condition

Each *active* neuron ( $a_j \neq 0$ ) satisfies  $R_j := \int_{b_j}^1 (f - f^*) dx \approx 0$  at stationarity. Verified numerically at every converged run; connects to the theory of OP 4.1.

# The Three Open Problems

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## OP 4.1: Cluster Count

$$\lim_{m \rightarrow \infty} C(m, f^*) = k$$

Does the number of bias clusters converge to the number of sign-changing inflection points of  $f^*$ ?

## OP 4.2: Higher Dimensions

For  $f^* : [-1, 1]^2 \rightarrow \mathbb{R}$ , does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

## OP 4.3: Pruning Bound

After collapse, is there a provable  $L^2$  error bound when pruning to  $k$  neurons?

$$\|\tilde{f} - f\|_{L^2} \leq \delta \cdot \sum_j |a_j|$$

Each open problem is introduced in detail immediately before its results.

# Experimental Design

## Nine target functions:

| $f^*(x)$       | $k$ | type |
|----------------|-----|------|
| $\sin(\pi x)$  | 1   | trig |
| $x^3$          | 1   | poly |
| $\sin(2\pi x)$ | 3   | trig |
| $x^5 - 3x^3$   | 3   | poly |
| $\sin(3\pi x)$ | 5   | trig |
| $\sin(4\pi x)$ | 7   | trig |
| $\sin(5\pi x)$ | 9   | trig |
| $\sin(6\pi x)$ | 11  | trig |
| $\sin(7\pi x)$ | 13  | trig |

Trig:  $k = 2n - 1$ . Poly: irregular inflection spacing.

## Main sweep (`simulate_parallel.py`):

- ▶  $m \in \{50, 100, 250, 500, 1000, 2000, 3500, 5000\}$ ,  
 $T = 500$
- ▶ **72 completed runs**, 9 targets; 70/72 fully stationary  
( $\max |\dot{a}_j| < 0.01$ )

## Supporting experiments:

| Experiment                                | Runs | Purpose             |
|-------------------------------------------|------|---------------------|
| <code>simulate_discrete.py</code>         | 52   | Discrete GD vs. ODE |
| <code>lottery_ticket_experiment.py</code> | 52   | OP 4.1.4 (survival) |
| <code>verify_pruning.py</code>            | 72   | OP 4.3 bound check  |
| <code>collapse_v2.py</code>               | 9    | OP 4.2, 2D targets  |
| <code>instability_test.py</code>          | 22   | Stability near $k$  |

# Open Problem 4.1: Cluster Count Conjecture

## Conjecture

$$\lim_{m \rightarrow \infty} C(m, f^*) = k,$$

where  $k := \#$  sign-changing inflection points of  $f^*$  in  $(-1, 1)$

$C(m, f^*)$  = number of distinct bias clusters at stationarity.  $k$  = number of sign-changing inflection points of  $f^*$ .

### Sub-problems we address:

- ▶ **4.1.1:** Can we prove  $C(m, f^*) \leq k$  independently of  $m$ ?
- ▶ **4.1.4:** Which neurons survive collapse to become cluster representatives?

## Why $k$ is the right target

For  $\sin(n\pi x)$ , the second derivative  $(f^*)'' = -n^2\pi^2 \sin(n\pi x)$  has exactly  $k = 2n - 1$  sign changes in  $(-1, 1)$ .

At stationarity, each active neuron satisfies

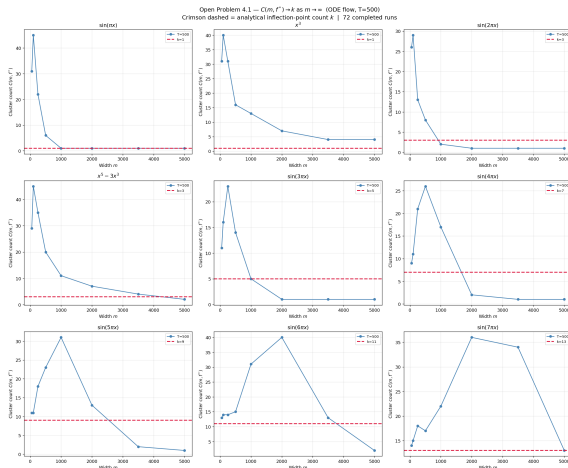
$$R_j := \int_{b_j}^1 (f - f^*) dx = 0.$$

A  $k$ -cluster arrangement satisfies this condition at the inflection locations.

## What makes this hard

Above- $k$  and below- $k$  stationary states both exist numerically. The conjecture requires showing gradient flow selects  $C = k$  from generic initializations.

# OP 4.1: $C(m, f^*)$ Across All 9 Targets



$C(m, f^*)$  vs.  $m$  at  $T = 500$  (72 runs, 9 targets). **Crimson dashed line =  $k$ .** All panels show rise-then-fall. Low- $k$  trig targets reach  $C = k$ ;  $\sin(7\pi x)$  ( $k = 13$ ) reaches  $C = 13$  at  $m = 5000$  for the first time (stationarity unconfirmed).

## OP 4.1: The Rise-Then-Fall Dynamic

### Universal pattern across all 9 targets:

1. **Rise:** Small  $m$  cannot populate all  $k$  inflection attractors simultaneously.  $C$  grows above  $k$ .
2. **Peak:**  $C$  reaches a maximum scaling with  $k$  (up to  $C = 40$  for  $k = 11$ ).
3. **Fall:** With sufficient redundancy, collapse proceeds and  $C$  decreases toward  $k$ .

#### Key observation

The peak of  $C(m)$  shifts to **larger**  $m$  as  $k$  grows. High- $k$  targets require much wider networks before the fall phase begins.

### Selected $C(m)$ at $T = 500$ :

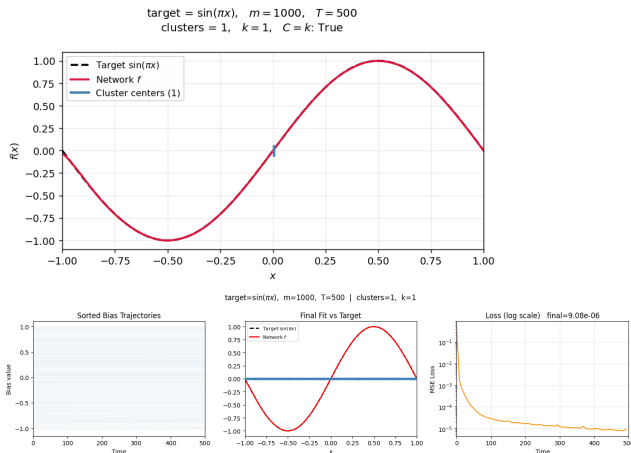
| Target         | $k$ | $m=1000$ | $m=2000$ | $m=5000$ |
|----------------|-----|----------|----------|----------|
| $\sin(\pi x)$  | 1   | <b>1</b> | 1        | 1        |
| $\sin(3\pi x)$ | 5   | <b>5</b> | †1       | †1       |
| $\sin(4\pi x)$ | 7   | 17       | 2        | †1       |
| $\sin(7\pi x)$ | 13  | 22       | 36       | ‡13      |
| $x^5 - 3x^3$   | 3   | 11       | 7        | 2        |
| $x^3$          | 1   | 13       | 7        | 4        |

† Cluster structure absent or too dense for fixed threshold to detect.

‡  $C = k$  but not fully stationary ( $\max |\dot{a}_j| = 0.034$ ).



## OP 4.1 Confirmation: $\sin(\pi x)$ , $k = 1$



$\sin(\pi x)$ ,  $m = 1000$ ,  $k = 1$

**Top:** Clean final fit vs. target. Blue ticks = single cluster center.

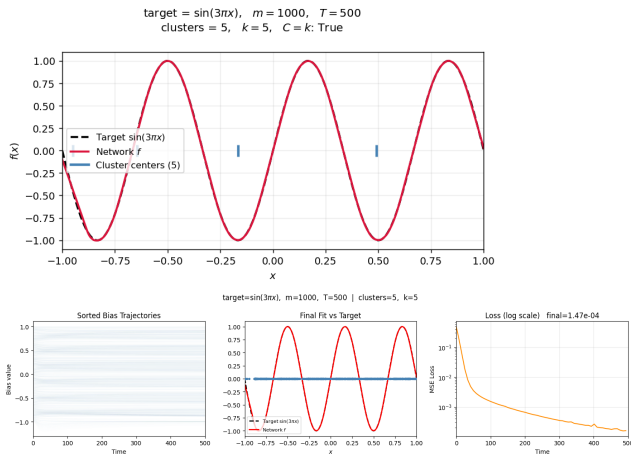
**Bottom:** Bias trajectories (all 1000 collapse to **one location**), final fit, and MSE loss on log scale.

### Strongest confirmation

$C = 1 = k$  for *all*  $m \geq 1000$  through  $m = 5000$ , fully stationary ( $\max |\dot{a}_j| < 3.6 \times 10^{-3}$ ).

Only target with  $C = k$  confirmed robustly across the full  $m$  sweep.

## OP 4.1 Confirmation: $\sin(3\pi x)$ , $k = 5$

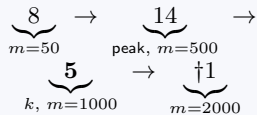


$\sin(3\pi x)$ ,  $m = 1000$ ,  $k = 5$

**Top:** Clean final fit. Five blue ticks mark the **5 bias clusters**, one per inflection point.

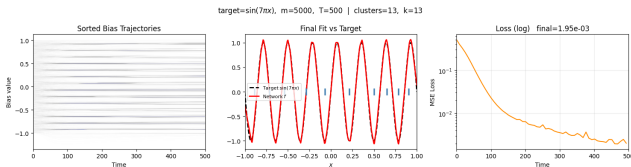
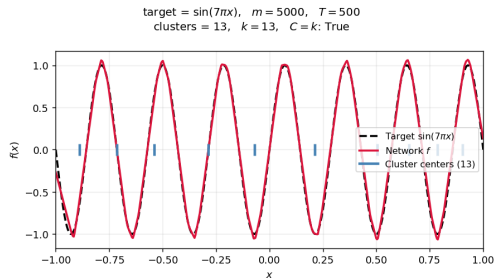
**Bottom:** Bias trajectories, final fit, MSE loss on log scale.

### Non-monotone path to $k$



Rises above  $k$ , reaches  $k = 5$ , then  
 $\uparrow$ cluster structure dissolves at larger  
 $m$ .

## OP 4.1 New at $m = 5000$ : $\sin(7\pi x)$ , $k = 13$



$\sin(7\pi x)$ ,  $m = 5000$ ,  $k = 13$

**Top:** Clean final fit. Thirteen blue ticks align with all 13 inflection points.

Most complex target tested. First time  $C = k = 13$  is reached.

### Result

$C = 13 = k$  at  $m = 5000$ .

Path:  $14 \rightarrow 22 \rightarrow 36 \rightarrow 34 \rightarrow 13$   
( $m = 50, 1000, 2000, 3500, 5000$ )

### Caveat

$\max |\dot{a}_j| = 0.034 > 0.01$ . May be **transient**; longer  $T$  needed to confirm.

## OP 4.1: The Below- $k$ Phenomenon

Three cases at verified stationary states with  $C < k$ . Re-run with 2 additional seeds each:

| Target         | $k$ | $m$  | seed 42 | seed 5 | seed 25 |
|----------------|-----|------|---------|--------|---------|
| $\sin(2\pi x)$ | 3   | 1000 | 2       | 1      | 1       |
| $\sin(4\pi x)$ | 7   | 2000 | 2       | 3      | 5       |
| $x^5 - 3x^3$   | 3   | 5000 | 2       | 2      | $3 = k$ |

$C$  **varies across seeds** in all three cases. No single below- $k$  value is consistent.

### Interpretation

$C$  varies across seeds  $\Rightarrow$  below- $k$  states are **initialization-dependent local minima**, not universal fixed points.

Gradient flow has multiple basins of attraction. The  $k$ -cluster basin exists but is not the only one.

$x^5 - 3x^3$  seed 25:  $C = k = 3$

First time  $x^5 - 3x^3$  reaches  $C = k$ . Shows the  $k$ -cluster attractor exists for polynomial targets, though harder to find.

## OP 4.1: Polynomial vs. Trigonometric Targets

Same  $k$ , very different convergence speed:

| Target         | $k$ | $m=1000$ | $m=2000$ | $m=5000$ |
|----------------|-----|----------|----------|----------|
| $\sin(\pi x)$  | 1   | 1        | 1        | 1        |
| $x^3$          | 1   | 13       | 7        | 4        |
| $\sin(2\pi x)$ | 3   | 2        | †1       | †1       |
| $x^5 - 3x^3$   | 3   | 11       | 7        | 2        |

$x^3$  reaches only  $C = 4$  at  $m = 5000$ , despite  $k = 1$ .

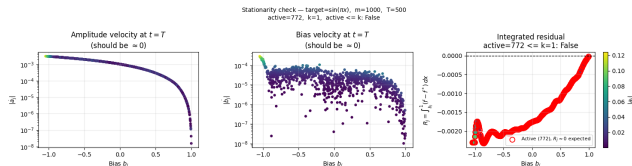
### Why inflection geometry matters

For  $\sin(n\pi x)$ , the  $k$  inflection points are *evenly spaced* across  $[-1, 1]$ , forming strong, symmetric attractors that neurons organize around early in training.

Polynomial targets have *irregular* spacing: weaker, asymmetric attractors. Neurons take far longer to find them.

**Implication:** a proof of OP 4.1 must account for attractor geometry, not just count.

## OP 4.1: Stationarity and Fixed-Point Verification



$\sin(\pi x)$ ,  $m = 1000$ ,  $T = 500$

Final  $(a_j, b_j)$  substituted back into the ODEs.

**Left:**  $|a_j| \approx 10^{-3} - 10^{-5}$

**Center:**  $|b_j| \approx 10^{-4} - 10^{-5}$

**Right:**  $R_j \approx 0$  for active neurons (red), confirming the theoretical fixed-point condition numerically.

### Discrete GD comparison

Discrete GD does *not* produce  $C \rightarrow k$  for any target except  $\sin(\pi x)$ . The conjecture is specific to continuous ODE gradient flow dynamics.

## Instability Near $k$ : Results (36/36 complete)

**Question:** Are above- $k$  states unstable? Does gradient flow dissolve extra clusters toward  $k$ ?

**22 qualifying runs:** 8 exact\_ $k$ , 8 above\_ $k$ , 6 below\_ $k$ .  
 $T_{\text{perturb}} = 1000$ .

| Category   | Status     | Finding                                                                         |
|------------|------------|---------------------------------------------------------------------------------|
| above_ $k$ | 8/8 done   | <b>No change.</b> $C$ stays fixed; ODE fixed points do not dissolve.            |
| below_ $k$ | 12/12 done | <b>No change.</b> Injected neuron amplitude stays at 0.01 and cannot grow.      |
| exact_ $k$ | 16/16 done | <b>All returned to <math>k</math>.</b> Exact- $k$ states are stable attractors. |

### above\_ $k$ : stable fixed points

Continuing ODE for  $T = 1000$ :  $C$  does **not** decrease. Above- $k$  states are *stable* fixed points, not metastable.

### below\_ $k$ : injection-resistant

Injected neurons gain **zero amplitude**. ODE provides no restoring force toward  $k$ . Supports: below- $k$  states are true fixed points.

### Implication for OP 4.1.1

$C \rightarrow k$  cannot rely on above- $k$  instability. Proof must use *initialization geometry*, not gradient flow instability.

## Open Problem 4.2: Higher-Dimensional Collapse

### Question

For a structured target  $f^* : [-1, 1]^2 \rightarrow \mathbb{R}$ , does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

**2D network:**

$$f(x) = \sum_{j=1}^m a_j \sigma(w_j^\top x + b_j), \quad x \in [-1, 1]^2$$

Each neuron defines an oriented hyperplane (line in 2D). Scale-invariant:  $\theta_j = \angle(w_j)$ ,  $\beta_j = b_j / \|w_j\|$ .

**Collapse in 2D:** the  $(\theta_j, \beta_j)$  cloud concentrates onto a low-dimensional subset.

### Diagnostic

Angular entropy  $H(\{\theta_j\})$ : near 0 if collapsed; near  $\log(36) \approx 3.58$  if uniform. Effective neurons  $N_{\text{eff}} = e^H$ :  
 $N_{\text{eff}}=1$  full collapse;  $N_{\text{eff}} \approx 36$  uniform.

### Width sweep

$m \in \{64, 256, 512, 1024\}$ , 6000 epochs, Adam.

### Three targets studied

Ridge, Separable, and Radial.  
See next slide for expected vs. observed behavior.



## Open Problem 4.2: Target Functions

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### Ridge

$$\sin(2\pi u^\top x) + 0.5 \sin(4\pi u^\top x)$$

*Expected (small width):*  
concentration near  $\theta \approx 53^\circ$

*Observed:*  
concentration weakens as width increases

### Separable

$$\sin(2\pi x_1) + 0.3 \sin(2\pi x_2)$$

*Expected (small width):*  
two dominant direction families

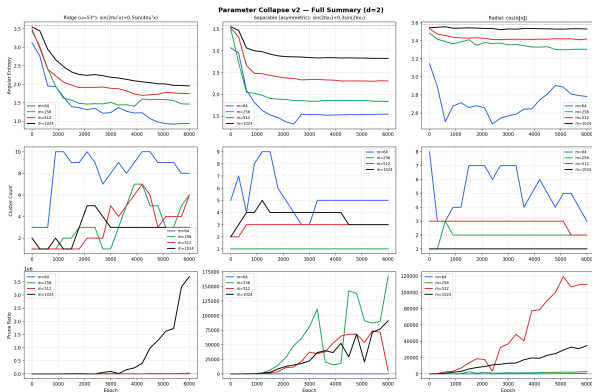
*Observed:*  
directional structure becomes diffuse at large width

### Radial (reference)

$$\cos(\pi \|x\|)$$

*Reference case:*  
approximately isotropic orientation distribution

## OP 4.2: Orientation Distributions



Entropy  $H$  / Eff. neurons  $N_{\text{eff}} = e^H$  at epoch 6000:

| Target    | $m=64$    | $m=256$   | $m=512$   | $m=1024$         |
|-----------|-----------|-----------|-----------|------------------|
| Ridge     | 0.94 / 3  | 1.46 / 4  | 1.74 / 6  | 1.95 / 7         |
| Separable | 1.55 / 5  | 1.84 / 6  | 2.31 / 10 | 2.83 / 17        |
| Radial    | 2.78 / 16 | 3.31 / 27 | 3.42 / 31 | <b>3.53 / 34</b> |

$N_{\text{eff}}=1$ : full collapse;  $N_{\text{eff}} \approx 36$ : uniform. Max  $H = \log(36) \approx 3.58$

### Open question

Why does directional concentration weaken as width grows? Observed collapse appears strongest in moderately overparameterized networks. The mechanism behind this trend remains unclear.

## OP 4.2: Key Findings

### What the data shows

- ▶ At small width, directional concentration reflects target geometry:  
ridge  $\rightarrow$  one direction family;  
separable  $\rightarrow$  two families;  
radial  $\rightarrow$  broadly distributed.
- ▶ Concentration decreases steadily as width grows.
- ▶ Networks increasingly fit targets using many distinct directions.

### Effective directional complexity

$N_{\text{eff}} = e^H$  measures how many distinct directions the network actively uses.

- ▶  $N_{\text{eff}}$  increases monotonically with width.
- ▶ Wider networks rely less on directional collapse.
- ▶ Radial  $m=1024$ :  $N_{\text{eff}} \approx 34$  of 36 — nearly uniform.

## Open Problem 4.3: Pruning Bound

### Bound (from the slides)

$$\|\tilde{f} - f\|_{L^2} \leq \delta \cdot \sum_{j=1}^m |a_j|$$

**Setup:** After gradient flow, bias clusters have formed. *Prune* by replacing each cluster with a single representative neuron, giving  $\tilde{f}$  with only  $k$  neurons.

$\delta$  = maximum intra-cluster diameter (how spread out the biases are within each cluster).

$\sum |a_j|$  = total weight mass of the full  $m$ -neuron network.

**Why it matters:** Provides a formal  $L^2$  certificate that collapsing from  $m$  neurons to  $k$  neurons is safe, bridging the collapse phenomenon and approximation theory.

### Why the bound holds

Within a cluster, all  $b_j$  differ by at most  $\delta$ , so  $\sigma(x - b_j) \approx \sigma(x - b_{\text{center}})$  up to an error  $\leq \delta$ . Summing over  $m$  neurons, errors accumulate at most as  $\delta \cdot \sum |a_j|$ .

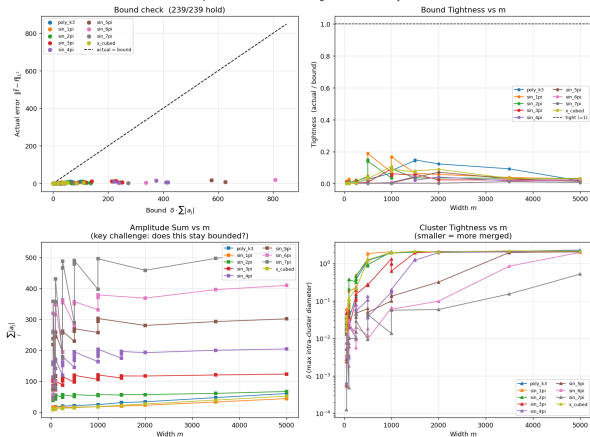
### The challenge

For the bound to be *non-vacuous* (goes to 0 as networks grow), we need  $\delta \cdot \sum |a_j| \rightarrow 0$ .

Our data shows  $\sum |a_j| \sim \sqrt{m}$ . So we need  $\delta \rightarrow 0$  fast enough, which must come from longer training  $T$ , not wider networks  $m$ .

# OP 4.3: Pruning Bound Verification

Open Problem 4.3 — Pruning Bound Summary



$$\text{Bound: } \|\tilde{f} - f\|_{L^2} \leq \delta \cdot \sum_j |a_j|$$

72 / 72 runs: bound holds

Tightness (actual / bound): 0.0002 to 0.148. Substantial slack in all cases.

Key obstacle:

- ▶  $\sum_j |a_j| \sim \sqrt{m}$  empirically
- ▶  $\delta$  does not shrink with  $m$  at  $T = 500$
- ▶ Bound loosens  $\sim \sqrt{m}$  while actual error plateaus

Path to non-vacuousness

$\delta \rightarrow 0$  requires  $T \rightarrow \infty$ , *not*  $m \rightarrow \infty$ .  
Study  $\delta(T)$  at fixed  $m$  to make the bound formally useful.

## OP 4.1.4 Lottery Ticket: Which Neurons Survive?

**Setup:** At  $t = 0$ , amplitudes  $a_j \sim \mathcal{N}(0, 0.01)$  are random. Only the bias placement  $b_j$  is structured. Neurons near an inflection point are *geometrically lucky*.

**Five conditions per (target,  $m$ ):** full, geometric, bias\_only, amp\_only, random\_k  
52 runs; 4 targets,  $m \in \{500, 1000, 1500\}$ .

### Claim 2 (performance): FALSE

Lucky  $k$  neurons alone achieve  $100\times$ – $50,000\times$  higher loss than the full network.  $k$  kinks cannot approximate a smooth target; full redundancy is necessary.

### Claim 1 (survival): bias position is key

**geometric**  $\approx$  **bias\_only** in every case (loss diff  $< 0.002$ ). Initial amplitudes carry *no information*.

**amp\_only** is the worst condition: good amplitudes with random biases performs worse than good biases with random amplitudes.

### Conclusion:

Bias position at  $t = 0$  is the sole informative quantity for collapse. The “winning ticket” is a good bias placement, not a good amplitude.

# Summary of Evidence

| Problem                      | Main finding                                                                                                                                                                                                       | Status                                                       |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| OP 4.1:<br>$C \rightarrow k$ | Confirmed for $\sin(\pi x)$ ( $k = 1$ ), all $m \geq 1000$ .<br>$\sin(7\pi x)$ : $C = 13 = k$ at $m = 5000$ (stationarity unconfirmed). Below- $k$ : <b>init-dependent local minima</b> (multiple seeds).          | <b>Supported</b> for $k = 1$ ;<br><b>open</b> for $k \geq 3$ |
| Instability<br>near $k$      | above- $k$ : stable fixed points (8/8, no dissolution).<br>below- $k$ : injection-resistant (12/12, zero amplitude growth).<br>exact- $k$ : all 16 returned to $k$ (stable attractors).                            | <b>Complete</b>                                              |
| OP 4.2:<br>2D collapse       | Directional concentration tracks target structure at small width. Decreases consistently with width ( $m \in \{64, 256, 512, 1024\}$ ). Effective directional complexity ( $N_{\text{eff}}$ ) increases with $m$ . | <b>Partial</b>                                               |
| OP 4.3:<br>Pruning bound     | 72/72 hold. $\sum  a_j  \sim \sqrt{m}$ . Non-vacuousness requires $\delta \rightarrow 0$ as $T \rightarrow \infty$ .                                                                                               | <b>Verified numerically</b> ;<br><b>open theoretically</b>   |
| OP 4.1.4:<br>Lottery ticket  | Bias position at $t = 0$ is sole informative quantity. Amplitude values irrelevant. Performance claim false.                                                                                                       | <b>Resolved</b>                                              |

# Open Questions and Next Steps

## OP 4.1 (open):

- ▶ A proof of  $C \rightarrow k$  must account for initialization geometry. Below- $k$  local minima exist, so the  $k$ -cluster basin must be shown to dominate at large  $m$
- ▶ **4.1.1:** above- $k$  states are *stable* fixed points, so instability alone cannot drive  $C \rightarrow k$
- ▶  $\sin(7\pi x)$ : Re-run at  $T = 1000$  to confirm stationarity of  $C = 13$

## OP 4.2 (open):

- ▶ Why does directional concentration weaken as width grows? The mechanism behind this trend remains unclear.

## OP 4.3, path to a formal proof:

- ▶ Study  $\delta(T)$  at fixed  $m$ : does  $\delta \rightarrow 0$  as  $T \rightarrow \infty$ ?
- ▶ Exponent  $\alpha$  in  $\sum_j |a_j| = O(m^\alpha)$ ; data suggests  $\alpha \approx 1/2$
- ▶ If  $\delta = O(T^{-\gamma})$  the bound becomes non-vacuous as  $T \rightarrow \infty$

### Immediate priorities

- (1)  $\sin(7\pi x)$  at  $T = 1000$  (confirm  $C = 13$ )
- (2)  $\delta(T)$  trajectory study
- (3) Theory for capacity-constrained collapse



# Conclusion

## What we showed:

- ▶  $C \rightarrow k$  confirmed for  $\sin(\pi x)$  ( $k = 1$ ),  $m \geq 1000$ .  
First high- $k$  evidence:  $\sin(7\pi x)$  reaches  $C = 13$  at  $m = 5000$ .
- ▶ **Rise-then-fall** is universal: peak scales with  $k$ ; inflection geometry sets the convergence rate.
- ▶ **Below- $k$** : 3 targets have  $C < k$  at genuinely stationary states, an open challenge for OP 4.1.
- ▶ **Instability**: Above- $k$  states are stable fixed points. Proof of  $C \rightarrow k$  must use initialization geometry.
- ▶ **OP 4.2**: Effective directional complexity grows with width; wider networks rely less on orientation collapse than narrower ones.
- ▶ **OP 4.3**: Bound holds  $72/72$ ;  $\delta \rightarrow 0$  with  $T$  is the path to a formal proof.
- ▶ **Lottery ticket**: bias position at  $t = 0$  is the sole informative quantity.

## Broader implication

Gradient flow is not only an optimizer; it is a **compression algorithm** that discovers the minimum-complexity representation of the target's intrinsic structure.

Inflection points of  $f^*$  act as **fixed-point attractors**: neurons near one survive; the rest dissolve.

A rigorous proof would explain generalization in overparameterized networks.

**Thank you**

*Supported by NSF RTG*